



Algorithms (10636314)
Dynamic Programming Project
Summer 2022/2023

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Section:	1 (9:30-11:00).



Part1: Divide & Conquer

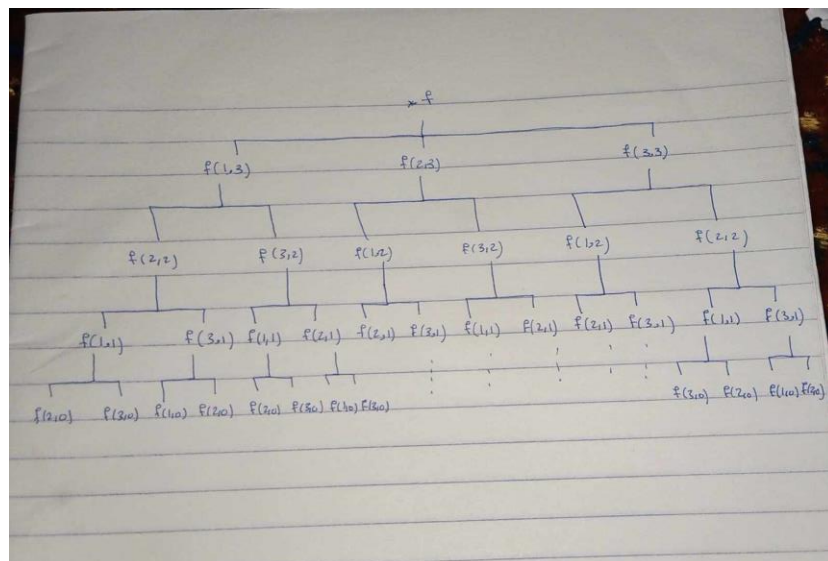
1- $\text{int } f_{\min}()$: return minimum amount of sugar.

2- $\text{int } f_{\min}(\text{int } i, \text{int } n)$

i : type of sweet.

n : number of days.

3- The recursion tree:



4- the recursive (divide and conquer) code:

```

3 usages
static int[][] sugarAmount;

2 usages
static int types[][] = {{2, 3},{1, 3},{1, 2}};

5 usages
public static int f(int i,int n){
    if(n<=0)
        return 0;
    int t1=f(types[i-1][0], n-1);
    int t2=f(types[i-1][1], n-1);
    return Math.min(t1,t2)+sugarAmount[i-1][n-1];
}

```

- | Type / Day | Day 0 | Day 1 | Day 2 | Day 3 |
|------------|-------|-------|-------|-------|
| | 0 | 0 | 0 | 0 |
| Type 1 | 0 | * | | |
| Type 2 | 0 | | ? | |
| Type 3 | 0 | * | | |

2- To determine the direction of movement within the table:

column.

need these-
 $f(1,1)$ $f(3,1)$

- H table

Days	Day1	Day2	Day3
X	X	X	X
X			
X			
X			



Fill F table

Dynamic programming code:

Fill H table

```
public static int dP(int days){
    for(int j=1;j<=days;j++){
        for(int i=1;i<=3;i++){
            int t1=F[types[i-1][0]][j-1];
            int t2=F[types[i-1][1]][j-1];
            F[i][j]=Math.min(t1,t2)+sugarAmount[i-1][j-1];
            if(t1<t2)
                H[i][j]=types[i-1][0];
            else
                H[i][j]=types[i-1][1];
        }
    }
    int temp=Math.min(F[1][days],F[2][days]);
    return Math.min(temp,F[3][days]);
}
```

4- The code that will print the sequence of moves that give me the solution:

```
usage
public static void printSeq(int days, int type) {
    for (int day =1; day <=days; day++) {
        System.out.println("Day " + day + ": Eat sweet type " + type);
        type = H[type][day];
    }
}
```

```
"C:\Program Files\Java\jdk-17.0.1\bin\java.exe"
3
3 6 9
6 3 5
8 2 6
10
Day 1: Eat sweet type 1
Day 2: Eat sweet type 3
Day 3: Eat sweet type 2
```